

Topic 8 – Coordination, Response and Gene Technology

Candidates will be assessed on their ability to:

8.1	know the structure and function of sensory, relay and motor neurones, including Schwann cells and myelination
8.2	understand how the nervous system of organisms can cause effectors to respond to a stimulus
8.3	know the structure and function of a spinal reflex arc, including grey matter and white matter of the spinal cord
8.4	understand how a nerve impulse (action potential) is conducted along an axon, including changes in membrane permeability to sodium and potassium ions
8.5	understand the role of myelination in saltatory conduction
8.6	(i) know the structure and function of synapses in nerve impulse transmission, including the role of neurotransmitters and acetylcholine (ii) understand how the pupil dilates and contracts
8.7	understand how the effects of drugs can be caused by their influence on nerve impulse transmission, illustrated by nicotine, lidocaine and cobra venom alpha toxin, the use of L-DOPA in the treatment of Parkinson's disease and the action of MDMA (ecstasy)
8.8	understand how the nervous systems of organisms can detect stimuli with reference to rods in the retina of mammals, the roles of rhodopsin, opsin, retinal, sodium ions, cation channels and hyperpolarisation of rod cells in forming action potentials in the optic neurones
8.9	understand what is meant by the term <i>habituation</i>
RECOMMENDED ADDITIONAL PRACTICAL	
Investigate habituation to a stimulus.	
8.10	know that the mammalian nervous system consists of the central and peripheral nervous systems
8.11	understand how phytochrome, auxin (IAA) and gibberellins bring about responses in plants, including their effects on transcription
8.12	CORE PRACTICAL 18 Investigate the production of amylase in germinating cereal grains.
8.13	understand how coordination in animals is brought about through nervous and hormonal control
8.14	know the location and main functions of the cerebral hemispheres, hypothalamus, pituitary gland, cerebellum and medulla oblongata of the human brain
8.15	understand how magnetic resonance imaging (MRI), functional magnetic resonance imaging (fMRI), positron emission tomography (PET) and computed tomography (CT) are used in medical diagnosis and the investigation of brain structure and function

8.16	understand how imbalances in certain naturally-occurring brain chemicals can contribute to ill health, including dopamine in Parkinson's disease and serotonin in depression, and to the development of new drugs
8.17	know how drugs can be produced using genetically modified organisms (plants, animals and microorganisms)
8.18	understand how recombinant DNA can be produced, including the roles of restriction endonucleases and DNA ligase
8.19	understand how recombinant DNA can be inserted into other cells
8.20	know how microarrays can be used to identify active genes
8.21	understand what is meant by the term <i>bioinformatics</i>
8.22	understand the risks and benefits associated with the use of genetically modified organisms